
Template Answers for Mid-Term for MT 3154 Graph Theory

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1. (a) Prove that a connected graph G on n vertices can be contracted to a single vertex using exactly $n - 1$ edge contractions. 2

[1] Since G is connected, it contains a spanning tree T . The tree T has n vertices and hence exactly $n - 1$ edges.

[0.5] Contract the edges of T , one at a time. Contracting an edge of the current image of T reduces its number of vertices by exactly one and keeps it connected.

After all $n - 1$ edges of T have been contracted, all vertices of G have been identified into a single vertex.

[0.5] Every edge contraction reduces the number of vertices by exactly one. Thus, to reduce the number of vertices from n to 1, exactly $n - 1$ edge contractions are necessary.

[0.5] We prove the statement by induction on $n = |V(G)|$.

If $n = 1$, there is nothing to contract, and $0 = n - 1$. Thus the assertion holds for $n = 1$.

[0.5] Now let $n \geq 2$, and assume as the induction hypothesis that every connected graph on $n - 1$ vertices can be contracted to a single vertex using exactly $n - 2$ edge contractions.

Since G is connected and has at least two vertices, it contains an edge $e = xy$.

[0.5] Contract the edge e . The graph G/e is connected and has $n - 1$ vertices. Hence, by the induction hypothesis, G/e can be contracted to a single vertex using exactly $(n - 1) - 1 = n - 2$ further edge contractions.

Including the contraction of e , this gives $1 + (n - 2) = n - 1$ edge contractions in total.

[0.5] Finally, every edge contraction reduces the number of vertices by exactly one. Therefore, reducing a graph from n vertices to one vertex requires at least $n - 1$ edge contractions.

Hence G can be contracted to a single vertex using exactly $n - 1$ edge contractions.

- (b) Prove that if G is a non-trivial graph, then $\kappa(G) \leq \lambda(G)$. Here, $\kappa(G)$ and $\lambda(G)$ are the greatest integers k and ℓ such that G is k -connected and ℓ -edge-connected, respectively. 3

[1] Put $\lambda(G) =: \ell$. Let F be a set of ℓ edges such that $G - F$ is disconnected. We may choose F minimal as an edge-separating set.

Suppose there is a vertex v that is not incident with an edge of F . Let C be the component of $G - F$ containing v . The vertices of C that are incident with an edge of F separate v from $G - C$.

By the minimality of F , no edge of F has both ends in C . Hence there are at most $|F|$ such vertices. Therefore $\kappa(G) \leq |F| = \lambda(G)$.

- [1] Suppose now that every vertex is incident with an edge of F . Let v be any vertex, and let C be the component of $G - F$ containing v .

Every neighbour w of v with $vw \notin F$ lies in C . Since every vertex is incident with F , such a vertex w is incident with an edge of F ; by the minimality of F , these edges can be chosen distinctly. Together with the edges of F incident with v , this gives $d_G(v) \leq |F|$.

- [0.5] If $\{v\} \cup N(v) \neq V(G)$, then $N(v)$ separates v from some other vertex, and hence $\kappa(G) \leq d_G(v) \leq |F| = \lambda(G)$.

- [0.5] If $\{v\} \cup N(v) = V(G)$ for every v , then G is complete. For a complete graph K_n , $\kappa(K_n) = \lambda(K_n) = n - 1$. Thus in all cases, $\kappa(G) \leq \lambda(G)$.

- (c) A set S of vertices in a graph G is called an *independent set* if no two vertices in S are adjacent. Define $\alpha(G)$ as the maximum cardinality of an independent set in G , and $\Delta(G)$ as the maximum degree of G . Prove that $\alpha(G) \geq \frac{|V(G)|}{\Delta(G)+1}$.

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- [1] Let S be a maximal independent set of G . Every vertex of $V(G) - S$ has a neighbour in S ; otherwise that vertex could be added to S , contradicting the maximality of S . Therefore

$$V(G) \subseteq \bigcup_{v \in S} (\{v\} \cup N(v)).$$

- [2] For every $v \in S$,

$$|\{v\} \cup N(v)| \leq \Delta(G) + 1.$$

Hence

$$|V(G)| \leq \sum_{v \in S} (1 + d(v)) \leq |S|(\Delta(G) + 1).$$

Thus

$$|S| \geq \frac{|V(G)|}{\Delta(G) + 1}.$$

Since $\alpha(G) \geq |S|$, we obtain

$$\alpha(G) \geq \frac{|V(G)|}{\Delta(G) + 1}.$$

2. (a) (1+3) Consider a bipartite graph G and, for every vertex $v \in V(G)$, a linear ordering $\leq_v$ on $E(v)$, the set of edges incident with v .

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1. Define stable matching in a bipartite graph.
2. Describe the Gale–Shapley algorithm (and the necessary terms used in it) to compute a stable matching.

- [1] Let the bipartition of G be $\{A, B\}$. A matching M is *stable* if for every edge $e \in E(G) \setminus M$, there is an edge $f \in M$ such that e and f have a common endvertex v and $e <_v f$.

- [0.5] Given a matching M , call $a \in A$ *acceptable* to $b \in B$ if $ab \notin M$ and every edge $f \in M$ incident with b , if such an edge exists, satisfies $f <_b ab$. Thus b prefers ab to its present matching edge.

- [0.5] Call $a \in A$ *happy* with M if a is unmatched, or its matching edge $f \in M$ satisfies $f >_a ab$ for every edge ab for which a is acceptable to b .

- [1] Start with the empty matching. All vertices of A are then happy.

Given a matching M for which all vertices of A are happy, choose an unmatched vertex $a \in A$ that is acceptable to some $b \in B$. Among all such edges ab , choose the a -maximal one. Add ab to M , and discard from M any other edge incident with b .

- [0.5] Repeat this operation. If no unmatched vertex of A is acceptable to any of its neighbours in B , terminate.
- [0.5] Whenever a vertex $b \in B$ changes its matching edge, it receives an edge it prefers to its previous one. This can happen only finitely many times. Hence the procedure terminates.

(b) (2+2) Consider a simple planar graph G .

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1. Prove that G contains a vertex of degree at most 5.
2. Suppose G contains a cycle and every cycle in it has at least 17 vertices. Find a better upper bound on the number of edges in G , in terms of the number of vertices of G , than the one implied merely by planarity.

[1] Write $n := |V(G)|$ and $m := |E(G)|$. For $n \leq 2$ the assertion is immediate. Assume $n \geq 3$. A simple planar graph satisfies $m \leq 3n - 6$.

[1] Hence $\sum_{v \in V(G)} d(v) = 2m \leq 6n - 12 < 6n$. Thus the average degree is less than 6, so at least one vertex has degree at most 5.

(2)

[1] Fix a plane drawing, and let f be its number of faces. Since every cycle has length at least 17, every facial boundary walk has length at least 17. Since every edge occurs twice among the facial boundary walks, $2m \geq 17f$. Thus $f \leq \frac{2m}{17}$.

[1] Euler's formula gives $n - m + f = 2$. Therefore

$$2 = n - m + f \leq n - m + \frac{2m}{17} = n - \frac{15m}{17}.$$

Hence $m \leq \frac{17}{15}(n - 2)$, which is stronger than $m \leq 3n - 6$.

3. (a) Consider a bipartite graph G with bipartition A, B . Let M, M' be matchings of G such that M saturates $S \subseteq A$ and M' saturates $T \subseteq B$. Prove that there exists a matching of G that saturates $S \cup T$.

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[1] Consider the graph H with vertex set $V(G)$ and edge set $M \cup M'$. Every vertex has degree at most 2 in H . Hence every non-trivial component of H is an alternating path or an alternating even cycle (with edges alternating between M and M').

[1] Every internal vertex of an alternating path, and every vertex of an alternating cycle, is incident with an edge of each matching.

Thus the only vertices of S that might fail to be saturated by M' are endpoints of alternating paths whose incident edge belongs to M .

Similarly, the only vertices of T that might fail to be saturated by M are endpoints whose incident edge belongs to M' .

[1] Suppose that one component contained both $s \in S$ unmatched by M' and $t \in T$ unmatched by M .

Then $s \in A$ and $t \in B$ are the two ends of an alternating path, as $S \subset A$ and $T \subset B$. Its first edge at s lies in M , while its last edge at t lies in M' . Hence this alternating path has even length. But a path from A to B in a bipartite graph has odd length, a contradiction.

[1] For each component C of H :

- if C contains a vertex of S not saturated by M' , take the M -edges of C ;
- if C contains a vertex of T not saturated by M , take the M' -edges of C ;
- otherwise take either set of alternating edges.

By the previous arguments, the above instructions never conflict, i.e., we can't have a component C of H where it there is a vertex of S not saturated by M' and a vertex of T not saturated by M .

The union of the chosen edges over all components is a matching, and it saturates every vertex of $S \cup T$.

- (b) Prove that every 3-connected graph $G \neq K^4$ has an edge e such that G/e is again 3-connected. 4

[1] Suppose, for a contradiction, that G/e is not 3-connected for every edge $e \in E(G)$. Let $xy \in E(G)$.

Then G/xy has a separator of at most two vertices.

Since G itself is 3-connected, the contracted vertex v_{xy} must belong to this separator, and the separator has exactly two vertices.

Hence there is a vertex $z \notin \{x, y\}$ such that $\{v_{xy}, z\}$ separates G/xy .

It follows that $T := \{x, y, z\}$ separates G .

[1] No proper subset of T separates G , since G is 3-connected.

Therefore every vertex of T has a neighbour in every component of $G - T$.

Choose an edge xy , a vertex z , and a component C of $G - \{x, y, z\}$ so that $|C|$ is as small as possible.

[1] Since z has a neighbour in every component of $G - T$, choose a neighbour $v \in C$ of z .

By our assumption, G/zv is not 3-connected.

Thus there is a vertex w such that $\{z, v, w\}$ separates G .

As before, every vertex of $\{z, v, w\}$ has a neighbour in every component of $G - \{z, v, w\}$.

[1] Since x and y are adjacent, there is a component D of $G - \{z, v, w\}$ such that $D \cap \{x, y\} = \emptyset$.

The vertex v has a neighbour in D . Since $v \in C$, and D avoids x, y, z , this neighbour lies in C . Thus $D \cap C \neq \emptyset$.

Because D is connected and avoids x, y, z , while C is a component of $G - \{x, y, z\}$, we have $D \subseteq C$. Moreover $v \in C$ but $v \notin D$, so $D \subsetneq C$.

This contradicts the choice of C with minimum cardinality. Therefore some edge e of G satisfies that G/e is again 3-connected.

4. (a) Without using Kuratowski's theorem, prove that a graph contains K^5 or $K_{3,3}$ as a minor if and only if it contains K^5 or $K_{3,3}$ as a topological minor. 6

[1] Recall the two relevant definitions.

A graph H is a *minor* of G if we can associate with every vertex $x \in V(H)$ a connected subgraph V_x of G , called a *branch set*, such that the branch sets are pairwise vertex-disjoint and, whenever $xy \in E(H)$, there is an edge of G joining V_x to V_y .

A graph H is a *topological minor* of G if G contains a subdivision of H .

Hence, every topological minor is a minor.

[0.5] If $K_{3,3} \preceq G$, then, since every vertex of $K_{3,3}$ has degree 3, a $K_{3,3}$ -minor model can be converted into a subdivision of $K_{3,3}$. Thus $K_{3,3} \preceq_{\text{top}} G$.

[0.5] The only non-trivial implication that remains to prove is

$$K^5 \preceq G \implies (K^5 \preceq_{\text{top}} G \vee K_{3,3} \preceq_{\text{top}} G).$$

Assume, that $K^5 \preceq G$ and choose a K^5 -minor model $K \subseteq G$ that is minimal with respect to deleting vertices and edges. Let $\{V_x \mid x \in V(K^5)\}$ be its five branch sets.

By minimality, each branch set V_x induces a tree. Also, between any two distinct branch sets V_x and V_y there is exactly one edge.

[0.5] Fix $x \in V(K^5)$.

The branch set V_x is joined to the four other branch sets by four edges.

Let T_x be the tree obtained from the tree induced by V_x by adding these four edges and their other endpoints.

The tree T_x has exactly four leaves, namely the four endpoints lying in the other branch sets.

We now distinguish two cases.

[1] **Case 1: Every tree T_x is a subdivision of $K_{1,4}$.**

For every $x \in V(K^5)$, let c_x be the unique vertex of degree 4 in T_x .

Consider two distinct vertices $x, y \in V(K^5)$. Let $p_{xy} \in V_x$ and $p_{yx} \in V_y$ be the ends of the unique edge joining V_x and V_y .

Inside V_x , there is a path from c_x to p_{xy} , and inside V_y , there is a path from p_{yx} to c_y . Together with the edge $p_{xy}p_{yx}$, these form a c_x - c_y path.

[0.5] Since T_x is a subdivision of $K_{1,4}$, the four paths leaving c_x towards the four other branch sets are internally vertex-disjoint.

Thus the five vertices $\{c_x : x \in V(K^5)\}$ are the branch vertices of a subdivision of K^5 contained in G . Therefore $K^5 \preceq_{\text{top}} G$.

[1] **Case 2: There exists a tree T_x which is not a subdivision of $K_{1,4}$.**

In this case T_x has exactly two vertices of degree 3; call them u and v .

After suppressing all vertices of degree 2, the tree T_x therefore has the following form: u and v are joined, two of the four leaves are attached on the u -side, and the other two leaves are attached on the v -side.

Let the four vertices of $K^5 - x$ be a, b, c, d , labelled so that the connections from V_x to V_a, V_b are on the u -side and the connections to V_c, V_d are on the v -side.

Now perform the following contractions.

[0.5] Contract each of the four branch sets V_a, V_b, V_c, V_d to a single vertex, denoted respectively by a, b, c, d . Inside V_x , contract all appropriate paths while keeping the two vertices u and v distinct. The resulting minor has six vertices u, v, a, b, c, d . The four vertices a, b, c, d induce a K^4 , because every pair of the corresponding branch sets was adjacent in the original K^5 -model.

[0.5] Now consider the two sets $A = \{u, c, d\}$, and $B = \{v, a, b\}$. Every vertex of A is adjacent to every vertex of B : (i) u is adjacent to v, a, b ; (ii) c is adjacent to v, a, b ; and (iii) d is adjacent to v, a, b .

Hence the resulting graph contains a $K_{3,3}$ as a subgraph. Therefore $K_{3,3} \preceq G$.

Since $K_{3,3}$ has maximum degree 3, as observed above, every $K_{3,3}$ -minor is also a topological $K_{3,3}$ -minor. Thus $K_{3,3} \preceq_{\text{top}} G$.

In Case 1 we obtained $K^5 \preceq_{\text{top}} G$, and in Case 2 we obtained $K_{3,3} \preceq_{\text{top}} G$. Therefore $K^5 \preceq G \implies (K^5 \preceq_{\text{top}} G \vee K_{3,3} \preceq_{\text{top}} G)$.

(b) Given a vertex x and a set $U \subseteq V(G) \setminus \{x\}$, an (x, U) -fan of size k is a collection of k paths from x to distinct vertices of U , any two of which share only the vertex x . 5

Using Menger's theorem, prove that a graph G with $|V(G)| > k$ is k -connected if and only if, for every $x \in V(G)$ and every $U \subseteq V(G) \setminus \{x\}$ with $|U| \geq k$, there exists an (x, U) -fan of size k .

[0.5] Formal statement of Menger's theorem:

Let $G = (V, E)$ be a graph and let $A, B \subseteq V$. Then the minimum number of vertices separating A from B in G is equal to the maximum number of disjoint A - B paths in G .

- [1] ($\Rightarrow$) Suppose first that G is k -connected. Let $x \in V(G)$, and let $U \subseteq V(G) \setminus \{x\}$, with $|U| \geq k$.

Consider the graph $G - x$, and the two vertex sets $A := N_G(x)$ and $B := U$.

- [1] We claim that at least k vertices are required to separate A from B in $G - x$.

Suppose, to the contrary, that a set Z with $|Z| < k$ separates A from B in $G - x$. Since $|U| \geq k > |Z|$, there is a vertex $u \in U \setminus Z$.

If there were an x - u path in $G - Z$, then after deleting its initial vertex x , the remaining path would contain an $N_G(x)$ - U path in $(G - x) - Z$, contradicting the fact that Z separates A from B .

Thus Z separates x from u in G , contradicting the k -connectedness of G .

Hence every A - B separator in $G - x$ has at least k vertices.

- [0.5] By Menger's theorem, $G - x$ contains k disjoint A - B paths $P_1, \dots, P_k$.

Let $a_i \in A = N_G(x)$ be the initial vertex of P_i . Adding the edge xa_i to P_i , for every i , gives k paths from x to distinct vertices of U .

Since the paths $P_1, \dots, P_k$ are disjoint, these new paths have only x in common. Hence they form an (x, U) -fan of size k .

- [1] ($\Leftarrow$) Conversely, suppose that the stated fan property holds, but that G is not k -connected.

Since $|V(G)| > k$, there exist distinct vertices x, y and a set $Z \subseteq V(G) \setminus \{x, y\}$, and $|Z| < k$, such that Z separates x from y .

Choose a set $U \subseteq V(G) \setminus \{x\}$ of exactly k vertices such that $Z \cup \{y\} \subseteq U$.

- [1] By assumption, there is an (x, U) -fan of size k . Since $|U| = k$, its k paths end at all the vertices of U .

In particular, one of these paths, say P , ends at y . For every vertex $z \in Z$, a different path of the fan ends at z .

As two paths of the fan have only x in common, P cannot contain any vertex of Z . Hence P is an x - y path in $G - Z$, contradicting the fact that Z separates x from y .

Therefore G is k -connected.